Lab 8 – Flint Water

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Formatting Instructions



- Please submit your lab report as a **pdf** to Gradescope.
- When you upload to Gradescope, please **match pages** with the **question number**.
- Be sure that all **group members** are **added** in your submission to Gradescope (click view/edit group on the top right of the page once shown your final submission after matching pages).

Assignment Overview

- We'll be investigating some data on water samples taken from Flint, Michigan in 2016, during the water crisis. We have the following analytical goals for this lab:
    - Evaluate the extent to which lead levels are unsafe across a representative sample of homes in the area
    - Investigate whether letting the faucet "flush" for a period of time lowers lead levels
    - Determine whether lead levels vary across wards (i.e., regions of the city).

STEP 0

- **Pre-lab work:** Complete the pre-lab tutorials ("Customizing ggplot2" and "if else statements") for Lab 8 first: https://stat212-learnr.stat.illinois.edu/
- **Download** the flint.xlsx file to your computer and then **import** into your RStudio session.
    - On the pop-up screen when importing, **click the drop-down under "Ward" and change it to a character variable**. Then click "Import" on the bottom right.
    - *If using **RMarkdown***, put the following code after your import code to allow Ward to be read categorically. Flint$Ward = as.character(Flint$Ward)
- Create a new **R script** (or use the **RMarkdown file** if you are using that option)
- Remember to **library(tidyverse)** so that you can use the ggplot function.

Data Description

This data set includes lead content measurements taken from tap water across 300 homes in Flint, Michigan (of which 269 homes' measurements are included). Researchers collected 3 water samples from each household: the water at first draw (faucet turned on), water after running the faucet for 45 seconds, and water after running the faucet for 2 minutes. Lead content is measured in parts per billion (ppb). The spreadsheet is organized such that one water sample is the unit of observation; there are 3 units of observation per household.

As a point of reference, lead measurements **above 5ppb** are considered ***somewhat unhealthy* for regular consumption,** and lead measurements **above 15ppb** are considered ***dangerous* for regular consumption.**

Variables

SampleID: Household number. There are 269 households that provided data.

Zip_Code: Household's zip code

Ward: The regional zone that the household was in. A ward is like a precinct. https://www.cityofflint.com/city-of-flint-ward-map/

Time: The time point at which the water sample was taken: First draw, after 45 seconds, or after 2 minutes.

Lead_ppb: The lead concentration in the water sample, measured in parts per billion (ppb)

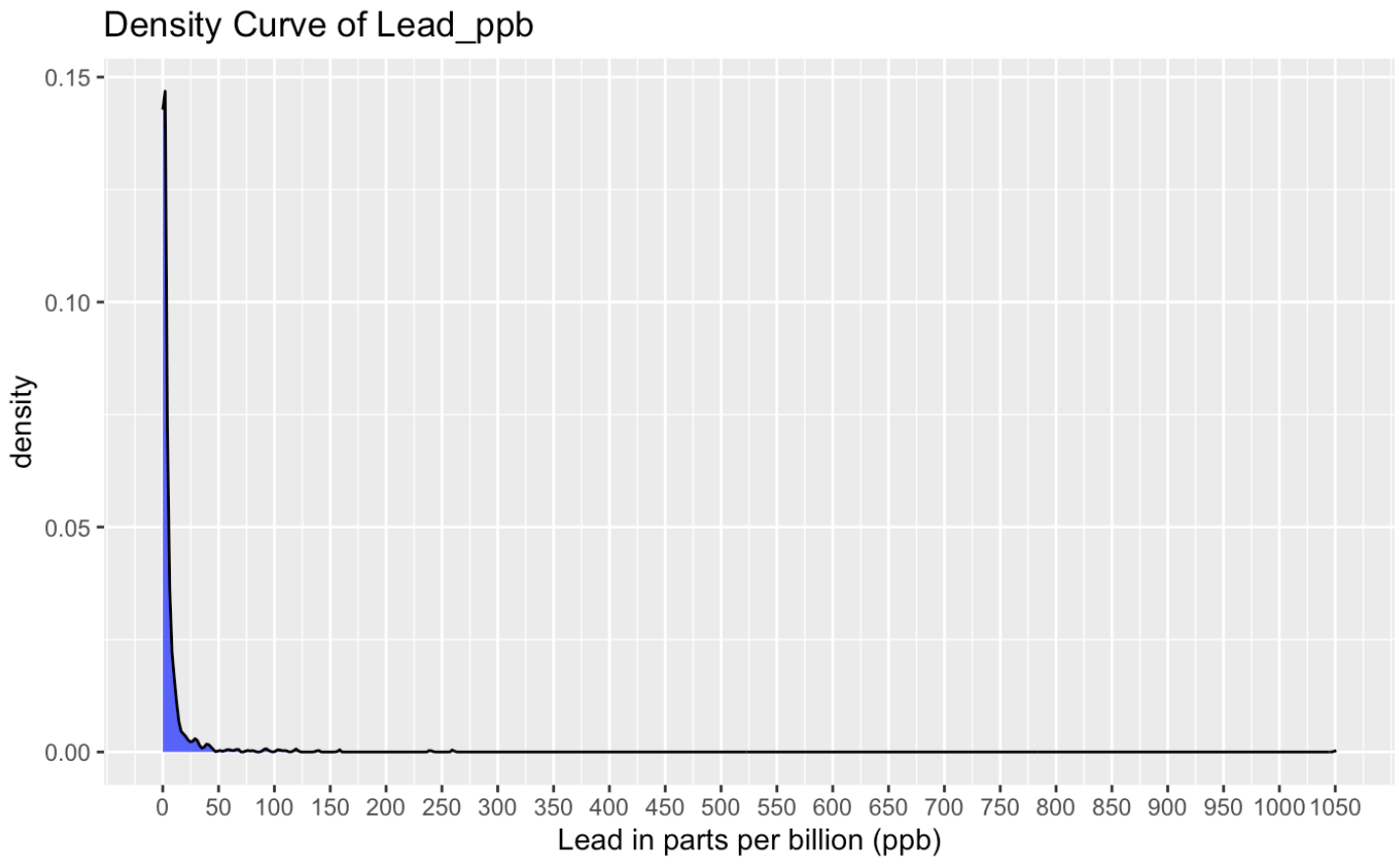
Question 1 (5pts): To get started, let's visualize our response variable. Create a density curve that plots the distribution of the `Lead_ppb` variable.

- Use a fill color
- Add a title, and label the x axis to state "Lead in parts per billion (ppb)"
- Scale the x axis to have more frequent tick marks than is shown by default (you be the judge!)

Include an image of your graph in your report. *Code is optional—partial credit for code attempt, but no graph.*

Report the numeric summary result of this variable (min, Q1, Q2, mean, Q3, max).

Briefly describe the shape of this distribution




```r
ggplot(Flint, aes(x = Lead_ppb)) +
  geom_density(fill = "blue", alpha = 0.7) +
  labs(title = "Density Curve of Lead_ppb",
       x = "Lead in parts per billion (ppb)") +
  scale_x_continuous(breaks = seq(0, max(Flint$Lead_ppb), by = 50))
```

```r
summary(Flint$Lead_ppb)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.031   0.509   1.843   8.208   5.155 1051.000
```

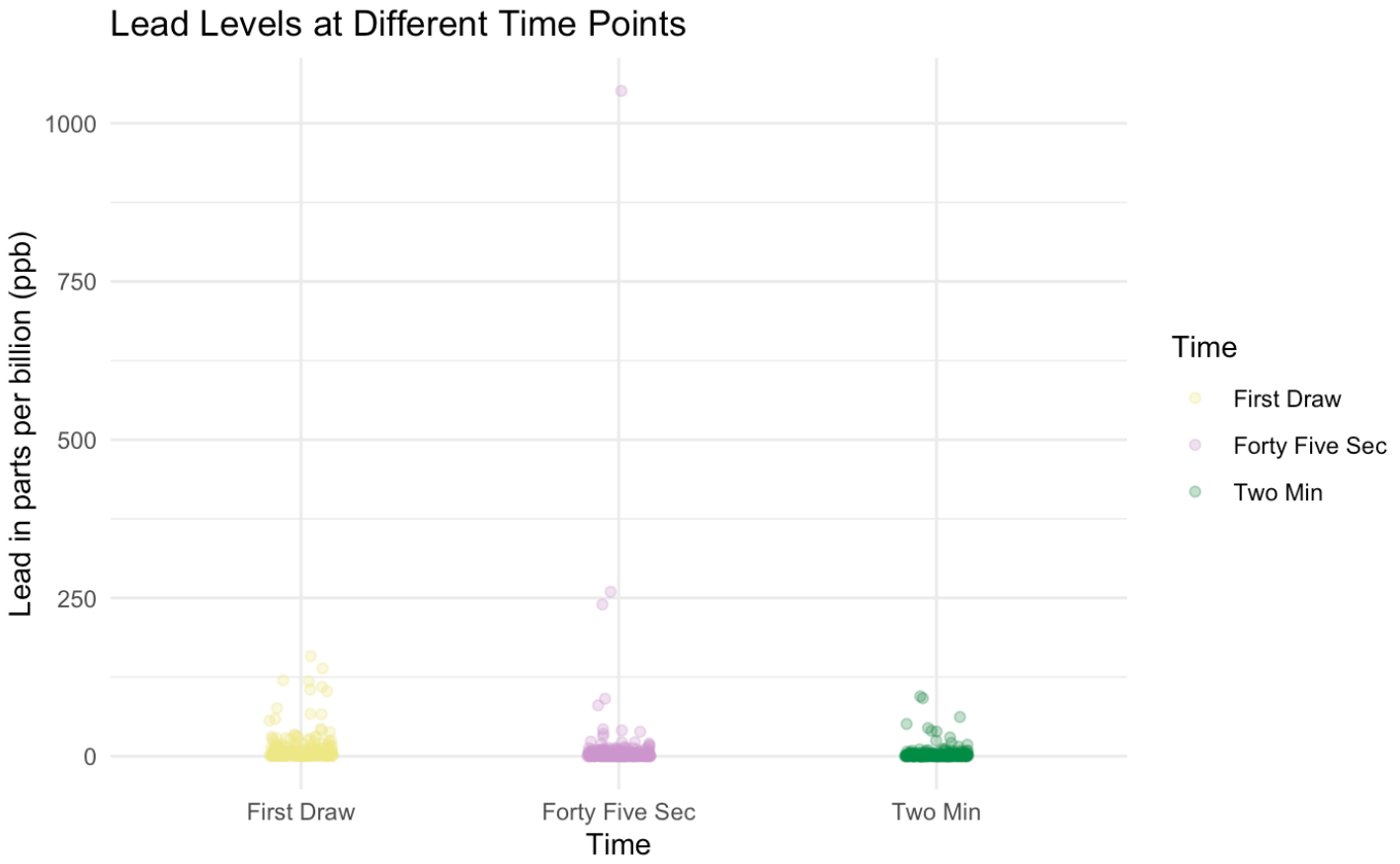
Min = 0.031, Q1 = 0.509, Q2 = 1.843, Mean = 8.208, Q3 = 5.155, Max = 1051

The shape of this distribution is highly right skewed.

Question 2 (5pts): Now, let's create jitter plots to visually compare the lead levels at the three different time points. Have each group be represented as a column of jittered points.

- Jitter your points with a width around 0.05 or 0.1
- Set an transparency (alpha) level between around 0.2 to 0.4
- Color each column differently, and customize the colors (manual, or with a color palette)
- Add a title, and label the lead axis to Lead in parts per billion (ppb)
- Use a plot theme

Report your graph in your report. *Code is optional—partial credit for code attempt, but no graph.*




```r
ggplot(Flint, aes(x = Time, y = Lead_ppb, color = Time)) +
  geom_jitter(width = 0.1, alpha = 0.3) +
  scale_color_manual(values = c("khaki2", "plum3", "springgreen4")) +
  labs(title = "Lead Levels at Different Time Points",
       x = "Time",
       y = "Lead in parts per billion (ppb)") +
  theme_minimal()
```

Question 3 (5pts): Using **a pipe**, let's create a summary table to compare lead concentrations, grouped by the timing of the water sample. Your summaries of lead concentrations should include the following:

- The median lead level, *rounded to (no more than) 3 decimal places*
- The mean lead level, *rounded to (no more than) 3 decimal places*
- Proportion of water samples with a lead level above 5ppb, *rounded to 3 decimal places*
- Proportion of water samples with a lead level above 15ppb, *rounded to 3 decimal places*

Report your **code** *and* report your **summary results** (you can copy/screenshot them as they are, or create a table in word if you prefer)

Notice that even though the 45 second group has a much lower median than the first draw group, it has almost the same mean. Consider your previous graph and briefly explain**: Which group do you think is producing the consistently highest lead measurements? What might be causing that spike in the mean for the other group?**

```r
Flint %>%
  group_by(Time) %>%
  summarize(
    medianLead = round(median(Lead_ppb), 3),
    meanLead = round(mean(Lead_ppb), 3),
    prop_5 = round(mean(Lead_ppb > 5), 3),
    prop_15 = round(mean(Lead_ppb > 15), 3)
  )
```

A tibble: 3 × 5

Time <chr>	medianLead <dbl>	meanLead <dbl>	prop_5 <dbl>	prop_15 <dbl>
First Draw	3.484	10.679	0.401	0.167
Forty Five Sec	1.400	10.266	0.245	0.063
Two Min	0.831	3.679	0.123	0.045

3 rows

The First Draw group consistently has higher median and mean lead levels compared to the Forty Five Sec group.

The close proximity of the mean lead levels between the two groups suggests that there might be some samples in the Forty Five Sec group with extremely high lead concentrations which are outliers that are influencing the mean.

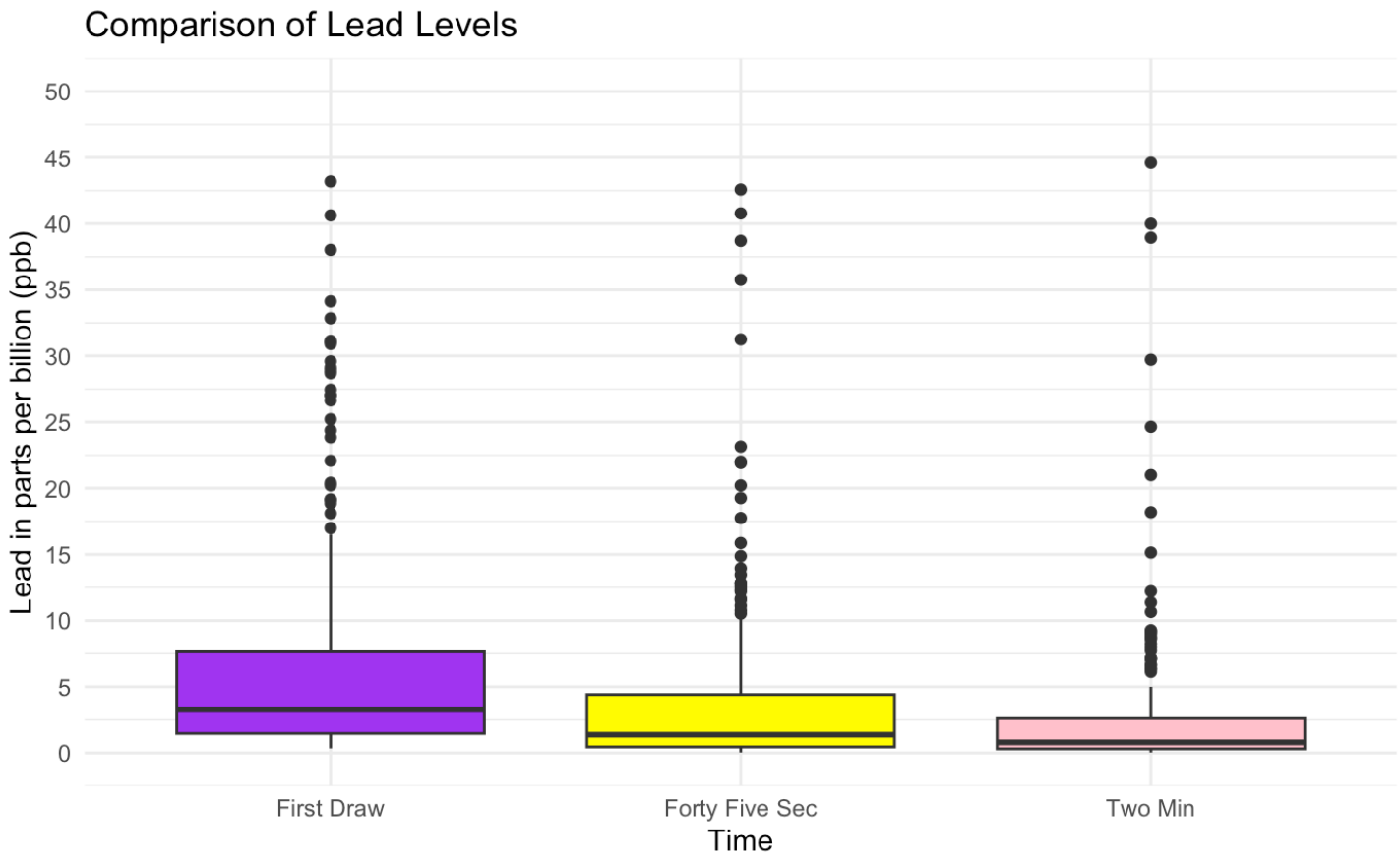
Question 4 (5pts): For sake of visualizing, let's narrow in on where most of the data is.

Create side by side **boxplots** of these same two variables and…

- Color each boxplot a different color. Use custom colors (or a color palette)
- Add a title, and label the lead axis to "Lead in parts per billion (ppb)"
- Use the scale function to have the y axis go in increments of 5 and have **limits** from 0 to 50
- Add a plot theme background
- *Remove* the color legend (you can do this using the theme() function).

Note the warning about "rows containing non-finite values" just means data points were excluded because they were outside the 0 to 50 range you limited your graph to—we did that on purpose!

Report your graph in your report. *Code is optional—partial credit for code attempt, but no graph.*




```r
ggplot(Flint, aes(x = Time, y = Lead_ppb, fill = Time)) +
  geom_boxplot() +
  scale_fill_manual(values = c("First Draw" = "purple", "Forty Five Sec" = "yellow", "Two Min" = "pink")) +
  labs(title = "Comparison of Lead Levels",
       y = "Lead in parts per billion (ppb)") +
  scale_y_continuous(breaks = seq(0, 50, by = 5), limits = c(0, 50)) +
  theme_minimal() +
  theme(legend.position = "none")
```

Question 5 (5pts): Let's now compare lead concentrations across Wards (precincts). Using **a pipe**, let's create a summary table to compare lead concentrations across Wards. Filter the data to *only include **first-draw** observations*. Your summaries of lead concentrations should include the following:

- The median lead level, *rounded to (no more than) 3 decimal places*
- The mean lead level, *rounded to (no more than) 3 decimal places*
- Proportion of water samples with a lead level above 5ppb, *rounded to 3 decimal places*
- Proportion of water samples with a lead level above 15ppb, *rounded to 3 decimal places*

Report your **code** *and* report your **summary results** (you can copy/screenshot them as they are, or create a table in word if you prefer)

```r
Flint %>%
  filter(Time == "First Draw") %>%
  group_by(Ward) %>%
  summarize(
    medianLead = round(median(Lead_ppb), 3),
    meanLead = round(mean(Lead_ppb), 3),
    prop_5 = round(mean(Lead_ppb > 5), 3),
    prop_15 = round(mean(Lead_ppb > 15), 3)
  )
```

A tibble: 9 × 5

Ward <chr>	medianLead <dbl>	meanLead <dbl>	prop_5 <dbl>	prop_15 <dbl>
1	2.362	4.066	0.129	0.065
2	2.952	9.820	0.417	0.167
3	3.442	12.518	0.318	0.136
4	1.966	8.623	0.314	0.114
5	2.377	10.296	0.389	0.222
6	3.679	20.111	0.484	0.290
7	5.223	11.142	0.526	0.237
8	5.402	14.838	0.552	0.207
9	3.832	6.617	0.439	0.098

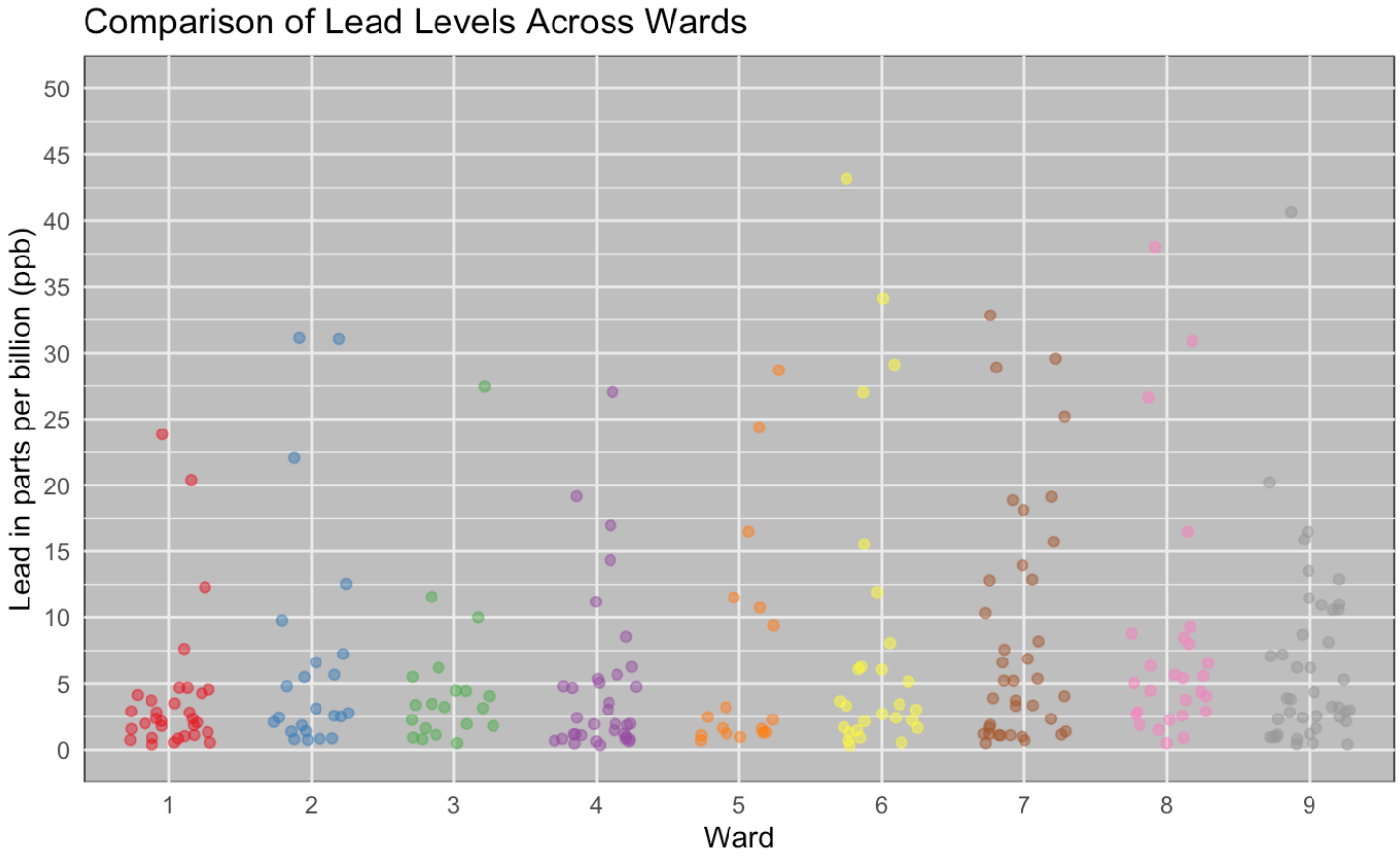
9 rows

Question 6 (5pts): Using a pipe, filter to only include first draw measurements. Then inside this pipe, build side-by-side jitter plots to compare lead ppb (at first draw) across each ward. *The reason we changed Ward to a factor variable was for this graph.*

- Color the points from each Ward a different color. **Use a color palette** for this one
- Add a title, and label the lead axis to "Lead in parts per billion (ppb)"
- Use the scale function to have the y axis go in increments of 5 and have **limits** from 0 to 50
- Add a plot theme background
- *Remove* the color legend (you can do this in the theme() function).

Report your graph in your report. *Code is optional—partial credit for code attempt, but no graph.*

Now that you have visualized and summarized the data, **do you notice much difference in lead concentration across wards?** If you had to pick, **which two or three wards seem to have the worst lead contamination?**




```r
Flint %>%
  filter(Time == "First Draw") %>%
  ggplot(aes(x = Ward, y = Lead_ppb, color = as.factor(Ward))) +
  geom_jitter(width = 0.3, alpha = 0.5) +
  scale_color_brewer(palette = "Set1") +
  labs(title = "Comparison of Lead Levels Across Wards",
       y = "Lead in parts per billion (ppb)") +
  scale_y_continuous(breaks = seq(0, 50, by = 5), limits = c(0, 50)) +
  theme_minimal() +
  theme(legend.position = "none", panel.background = element_rect(fill = "grey"))
```

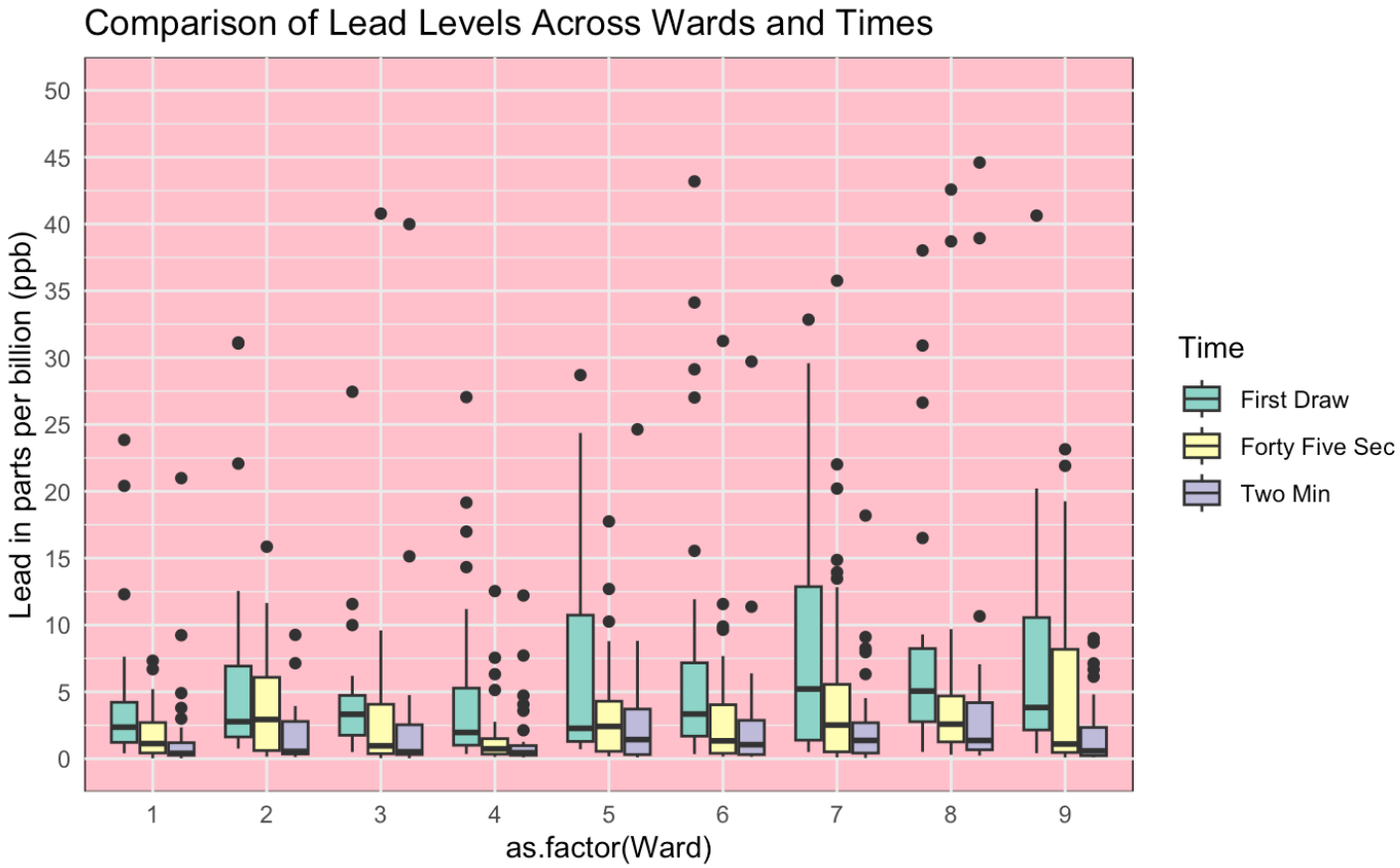
Question 7 (5pts): Now, let's make a graph that compares lead levels across Wards and across times together. Keep Ward on the x axis, and now map Time as the fill color.

- Use the boxplot geometry for this one
- Color (or fill) each boxplot a different color. Use a color palette for this one
- Add a title, and label the lead axis to "Lead in parts per billion (ppb)"
- Use the scale function to have the y axis go in increments of 5 and have limits from 0 to 50
- Add a plot theme background

Be sure that you make the viewer window nice and big to make everything clearly visible before copying into your report.

Report your graph in your report. *Code is optional—partial credit for code attempt, but no graph.*

If you were advising a resident in Flint who hadn't had their water tested for lead, **could you give them any data-based advice about how to maximize their safety if using water from their faucet?**



```r
Flint$Time <- factor(Flint$Time, levels = c("First Draw", "Forty Five Sec", "Two Min"))

ggplot(Flint, aes(x = as.factor(Ward), y = Lead_ppb, fill = Time)) +
  geom_boxplot() +
  scale_fill_brewer(palette = "Set3") +
  labs(title = "Comparison of Lead Levels Across Wards and Times",
       y = "Lead in parts per billion (ppb)") +
  scale_y_continuous(breaks = seq(0, 50, by = 5), limits = c(0, 50)) +
  theme_minimal() +
  theme(panel.background = element_rect(fill = "pink"))
```

Since the lead levels are consistently higher in the First Draw group compared to the 45 Sec and Two Min groups, residents may consider using water drawn first for tasks other than drinking or cooking, like watering flowers and flushing toilets.

Residents should flush the cold water tap for at least 5 minutes before using it for drinking or cooking, especially if the water has been sitting in the pipes for several hours. This helps reduce the risk of consuming water with elevated lead levels.